

Mike Hall

Principal Software Engineer

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TECHNICAL LEADERSHIP & PRODUCTION SYSTEMS MODERNIZATION

Principal Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines 20+ years across software engineering and platform architecture with OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, and cloud infrastructure.

AREAS OF FOCUS

- 01 Architecture & Modernization
- 02 Observability & AI-Augmented Engineering

EXPERIENCE

Development Manager — Founder Transition & Acquisition Handoff

EMR-Bear, a MedSuite company | Remote | May 2026 - August 2026

Engaged on a bounded 90-day interim assignment to conduct rapid architecture discovery, platform stabilization, and operational risk assessment for a multi-tenant healthcare SaaS platform serving 130+ clinics following a founder transition.

- Architecture Discovery & Operational Risk Assessment: Built a rapid 90-day system inventory and risk model spanning application codepaths, infrastructure dependencies, access boundaries, and operational risks.
- Production & Compliance Governance: Established release verification gates across the deployment pipeline of a platform servicing 130+ clinics, blocking unvalidated changes prior to HIPAA compliance review and business sign-off.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Served as Senior Technical IC and Software Architecture Lead for the Acquisition lane of a regulated financial enterprise, combining hands-on legacy modernization with production stability, OpenTelemetry observability, and cross-team technical direction.

- Technical Leadership & Team Consolidation: Progressed from Digital Origination Team Lead to Software Architect for the Acquisition Lane, merging two attrition-impacted teams into a unified, high-functioning engineering unit before converting to Associate Director, Staff Engineer upon corporate architecture track restructuring.
- ACQ Enablement Founding & Engineering Mandates: Founded and technically led the ACQ Enablement team to protect core engineering concerns from product feature pressure: driving platform stabilization, legacy modernization, system process mapping, architecture documentation, PII remediation, and fraud investigations.
- Observability & Enterprise Trace: Founded and led the OpenTelemetry Working Group (OTel WG) to establish the Enterprise Trace across high-volume customer acquisition flows, connecting a complex Rails distributed monolith, MuleSoft, and IBM mainframe backends to eliminate blind spots and align engineering around real-time service maps.
- AI-Augmented Systems Investigation: Introduced privacy-conscious local AI workflows for legacy code analysis and engineering support, expanding investigative capacity while keeping human review, security constraints, and source evidence explicit.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.

- Search Infrastructure Simplification: Replaced legacy external search clusters (Sphinx) with native MySQL full-text search and optimized relational indices, eliminating external service dependencies and reducing platform infrastructure footprint.
- Zero-Downtime Platform Upgrades: Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

SELECTED CURRENT WORK

Creator

Agent Tooling (MCP Servers & Agent Workflows) | Independent Project | 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- Bidirectional Knowledge Server: Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- Telemetry-Driven Diagnosis: Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- Context as a Budgeted Resource: Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

CORE CAPABILITIES & TECHNOLOGIES

ARCHITECTURE & MODERNIZATION

- Legacy Modernization & Architecture Discovery,
- Zero-Downtime Data Migration & Integrity,
- Incident Response & Root Cause Analysis,
- Distributed Systems & Asynchronous Processing

OBSERVABILITY & AI-AUGMENTED ENGINEERING

- OpenTelemetry & Observability Architecture,
- Model Context Protocol (MCP) Server Design,
- LLM Orchestration & Deterministic Evaluation,
- Platform Enablement & Developer Velocity

TECHNOLOGIES

- Ruby on Rails, PostgreSQL, Sidekiq & Redis, AWS, Kubernetes & Docker, CI/CD

SELECTED EARLIER EXPERIENCE

- **Senior Software Developer, BenchPrep** | March 2017 - February 2018 | Chicago, IL : Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote : Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Open-Source Transition Lead, Coderwall** | January 2014 - December 2014 | Remote : Hired as a contractor by founder Matt Deiters to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Engineering Learning & Development Business Partner, Groupon** | July 2011 - May 2013 | Chicago, IL : Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.

CAREER FOUNDATIONS

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.